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A sequential flow UNet for MRI brain tumor segmentation based on state-space-model

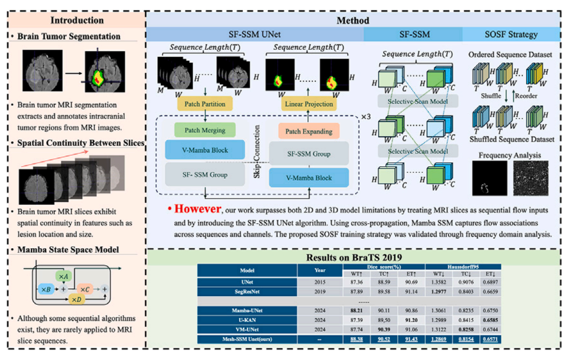
Jiacheng Lu, Hui Ding*, Qirun Huo, Kaiwen Wang, Xinyu Sun, Shiyu Zhang

College of Information Engineering, Capital Normal University, 105 West Third Ring Road North, Haidian District, Beijing, 100048, Beijing, China

HIGHLIGHTS

- We propose the SF-SSM UNet, a novel model for brain tumor MRI segmentation.
- The model serializes MRI slices and uses Mamba SSM to capture spatiotemporal correlations.
- We apply frequency-domain analysis to enhance sequential feature learning in MRI segmentation.
- On BraTS-2019 and MSD Task01, SF-SSM UNet reaches Dice >88 % and Hausdorff95 <1.3.

GRAPHICAL ABSTRACT



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ABSTRACT

The segmentation of brain tumors is one of the most extensively discussed challenges in medical image processing, aiming to differentiate various tumor regions from background regions in MRI scans. While UNet and its variants have demonstrated exceptional performance in 2D and 3D image segmentation, existing methods often neglect the temporal sequence associations between brain tumor MRI slices, which contain crucial information about lesion characteristics. To address this limitation, we propose the SF-SSM UNet (Sequential Flow State Space Model UNet), featuring the Sequential Flow Selective Scanning Module Group (SF-SSM Group). Our approach transcends traditional 2D and 3D models by treating MRI brain tumor slices as serialized image flow inputs. In the SF-SSM UNet, sequence flow is constructed through feature dimensional rearrangement, and Mamba SSM is applied across both sequence and channel dimensions, enabling the model to effectively capture temporal dependencies and feature interactions between slices. We further introduce the Shuffled-Ordered Sequential Flow Training Strategy (SOSF), a Fourier transform-based approach that enhances sequential feature learning and evaluates fine-grained performance using Frequency-domain Mean Error Ratio (FMER) and Frequency-domain Mean Error Number (FMEN). Experiments on BraTS-2019 show that SF-SSM UNet achieves Dice scores of 88.45, 90.55, and 91.44 for WT, TC, and ET regions, respectively, with corresponding Hausdorff95 distances all below 1.3. On MSD Task01, it further attains Dice scores of 89.26, 88.74, and 90.35. These results demonstrate clear performance gains across multiple benchmarks, establishing SF-SSM UNet as a state-of-the-art approach for sequential medical image segmentation.

* Corresponding author.
Email addresses: jchengl@foxmail.com (J. Lu), dhui@cnu.edu.cn (H. Ding), huoqirun@cnu.edu.cn (Q. Huo), kevincnu@163.com (K. Wang), xysun1224@163.com (X. Sun), sh1yuzh@163.com (S. Zhang).

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1. Introduction

Medical image segmentation [1] is a key step in visual information extraction and is particularly important for accurate diagnosis and analysis in brain tumor detection [2]. With the continuous development of computer-aided diagnosis (CAD) technologies, physicians can use computers to precisely diagnose diseases and tumors. However, the irregular boundaries, variable locations, and complex texture features of brain tumors, as well as issues such as uneven grayscale distribution, low interclass contrast, and blurred edges in MRI images, make achieving stable and accurate segmentation performance a widely discussed topic in current research.

With the widespread application of Convolutional Neural Networks (CNN) [3], UNet [4] and its various variants have achieved remarkable success in brain tumor image segmentation and demonstrated great potential. In recent years, researchers have also attempted to adopt cascaded approaches, which refine predictions in multiple stages to improve performance. For example, Cascaded UNet [46] and MambaBTS [47] have applied such methods in brain tumor segmentation, where a coarse segmentation is first performed to localize tumor regions, followed by fine segmentation within local areas.

Nevertheless, existing algorithms often overlook an important feature inherent in brain tumor MRI slices: sequential associations (as shown in Fig. 1). Typically, brain tumor image data are input into different models either as 3D images or as 2D slices. Due to the spatial continuity of features such as lesion location and size in MRI slices, 2D models are limited to the features of a single image, while 3D models [5–7], although capable of extracting some spatial features at the cost of high computational resources, still struggle to specifically process sequential continuity information.

Sequential models migrated from the field of natural language processing, such as LSTM-based methods [8], Transformer-based methods [9], and more recently segmentation methods based on the Mamba State Space Model (SSM) [10] [48–52], mostly construct sequences from image patches or windows [11–14], but fail to directly leverage the sequential continuity and lesion evolution characteristics inherent in MRI slices. Most existing algorithms pay little attention to the cross-slice temporal associations in MRI, and therefore, how to design an efficient segmentation algorithm for MRI brain tumor slice sequences that can fully model both temporal dependencies and channel correlations remains an urgent challenge and constitutes the primary motivation of this study.

The remainder of this paper is organized as follows. Section 1 introduces the background and motivation. Section 2 elaborates on the innovations and contributions of this work. Section 3 reviews related work, covering the development of medical image segmentation and the application of state-space models in sequential modeling. Section 4 details the proposed SF-SSM UNet method, including network architecture, core module design, and training strategy. Section 5 presents the experimental setup and results, including comparative and ablation studies on the BRATS-2019 and MSD Task01 datasets. Finally, Section 6 concludes the paper and discusses future research directions.

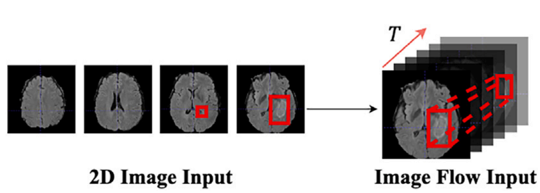

Fig. 1. Examples of 2D image input and image flow input of MRI slices. The image flow contains sequential associations of lesion scale and location, which is information that is difficult to capture with 2D segmentation.

2. Novelty

Given the inherent sequential characteristics of brain tumor magnetic resonance imaging (MRI) slices, there exists a promising research direction that transcends the limitations of conventional two-dimensional and three-dimensional segmentation models by leveraging advanced sequential algorithmic approaches. This study introduces the SF-SSM UNet algorithm, which conceptualizes MRI brain tumor slices as a coherent image sequence and strategically applies the State Space Model (Mamba SSM) across both sequential and channel dimensions. This approach allows the model to comprehensively capture the spatiotemporal dependencies and inter-channel correlations within the imaging data, enhancing its capacity to analyze and segment complex medical images. The primary scientific contributions of this research are as follows:

(1) We introduce the Sequential Flow State Space Model (SF-SSM) for image flow input, specifically designed to augment Mamba SSM's capability in extracting low-frequency semantic features. The module is engineered to enhance the model's ability to capture and interpret complex temporal and spatial associations inherent in brain tumor MRI slice sequences across both sequential and channel dimensions. The dual-sequence associative information is effectively integrated through a cross-flowing mechanism.

(2) We propose the SF-SSM Group (Sequential Flow State Space Model Group) and the SF-SSM UNet network, with SF-SSM as the foundational module. By serially connecting multiple SF-SSM blocks and incorporating a Layer Attention Mechanism, the model fully captures the temporal and spatial sequential information within the image flow, while layer-specific weighting enhances the precision of information extraction.

(3) We further explore the frequency-domain characteristics of sequential MRI slice flows and introduce two novel evaluation metrics, Frequency-domain Mean Error Ratio (FMER) and Frequency-domain Mean Error Number (FMEN), to quantify segmentation errors in high-frequency regions. These metrics provide a more rigorous assessment of model performance on fine-grained details. A simplified shuffled–ordered sequential training strategy is also adopted to enhance the robustness of frequency learning.

3. Related work

In the field of medical image analysis, medical image segmentation, particularly brain tumor segmentation, has become a focal point of research. In recent years, numerous studies have focused on incorporating sequential algorithms into segmentation models, with these algorithms predominantly being prominent techniques that have emerged in the field of Natural Language Processing (NLP).

3.1. Current state of the field

In medical image segmentation research, existing methods can be broadly divided into traditional approaches and deep learning-based approaches. Early brain tumor segmentation mainly relied on traditional image processing techniques, including thresholding, region growing, and clustering methods [15–18]. These approaches typically require manual intervention such as setting thresholds, selecting seed points, designing energy functions, or defining clustering rules, and are therefore highly dependent on human expertise [19]. When dealing with complex tumor morphologies and multimodal MRI images, such methods often fail to ensure accuracy and robustness.

Table 1 summarizes representative works of different categories along with their advantages and limitations. As shown in the table, compared with existing approaches, the proposed SF-SSM UNet can directly exploit the cross-slice temporal dependencies and channel correlations in MRI, thereby complementing the modeling capabilities of previous methods and addressing the limitation that prior work largely overlooked cross-slice sequential features.

Table 1
Comparison of representative methods for brain tumor segmentation.

Category	Representative methods	Characteristics (advantages / limitations)	Cross-slice sequential modeling
CNN-based	UNet [4], SLf-UNet [23]	+ Strong local feature extraction, efficient; widely adopted. – 2D models ignore inter-slice context; 3D models require high computational cost.	No
Transformer-based	TransUnet [24], UNETR [26], Swin UNETR [27]	+ Long-range dependency modeling, strong global context. – High computational and data demand; weak utilization of MRI sequential continuity.	No
Cascaded approaches	CasCaded UNet [46], MambaBTS [47]	+ Stage-wise refinement, improved local accuracy. – Complex pipeline; error accumulation; low efficiency.	No
SSM-based	Mamba-UNet [30], VM-UNet [42], UD-Mamba [44], H-VMUNet [45]	+ Linear complexity; global receptive field; efficient sequence modeling. – Mostly applied to patch/window sequences; not directly exploiting cross-slice continuity.	No
Proposed SF-SSM UNet	–	+ Directly models cross-slice sequential and channel dependencies. – Further exploration of the generalization ability across diverse lesion types and imaging modalities is still required.	Yes

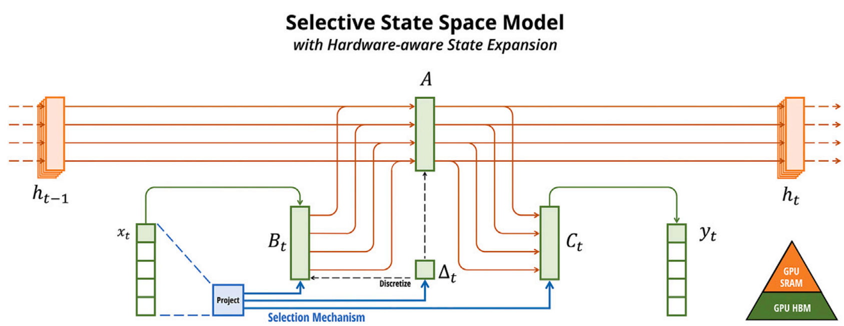

Fig. 2. Illustration of the state update computation in the Selective Scan State Space Model (S6) [10]. The three subfigures correspond to the progressive accumulation of these contributions, matching the mathematical formulation in Eq. (1).

(1) **CNN-based methods.** With the rise of Convolutional Neural Networks (CNNs), deep learning has gradually become the mainstream in medical image segmentation. U-Net, proposed by Ronneberger et al. in 2015 [4], is a seminal work in this field. Its symmetric encoder–decoder structure can efficiently capture multi-scale features and has become the foundation for subsequent research. Numerous variants based on U-Net have since been developed. For example, Med-SAM [20] incorporates domain-specific knowledge modules to address the limited generalization of the Segment Anything Model (SAM) [21] in medical scenarios, while SLf-UNet [23], proposed by Ding et al., introduces zFUP and sf modules to effectively fuse spatial and low-frequency features, further improving brain tumor segmentation performance. Although these CNN-based methods have achieved significant progress in 2D segmentation, their main limitation is that 2D models cannot exploit inter-slice contextual information, whereas 3D models incur high computational costs and still struggle to capture sequential dependencies.

(2) **Transformer-based methods.** In recent years, inspired by natural language processing models, researchers have introduced sequential modeling mechanisms into medical image segmentation tasks. Representative works include TransUnet [24], TransNorm [25], UNETR [26], and Swin UNETR [27], which combine U-Net with Transformer [9] to enhance the representation of long-range dependencies and global context. Furthermore, Roy et al. proposed MedNeXt [28], which integrates residual up-sampling and down-sampling blocks into a 3D encoder–decoder framework and employs UpKern to iteratively expand convolution kernels, thereby significantly improving segmentation performance even under limited data conditions. However, these methods typically require substantial data and computational resources, and their sequential modeling mainly focuses on spatial long-range dependencies rather than explicitly exploiting the temporal continuity of MRI slices.

(3) **SSM-based methods.** With the introduction of Mamba [10] at the end of 2023, State Space Models (SSMs) have gradually gained attention in visual segmentation tasks. Vision Mamba [29] and its derivatives have been applied to medical image segmentation, among which Mamba-UNet [30] is a representative work. In addition, recent approaches such as Mamba Sea [43], UD-Mamba [44], and H-VMUNet [45] leverage the selective scanning mechanism to achieve linear-complexity sequence modeling, maintaining a global receptive field while significantly reducing computational overhead, thus achieving superior performance in various segmentation tasks. However, most of these methods are still based on image patches or local window sequences and have not directly exploited the cross-slice sequential dependencies and lesion evolution characteristics inherent in MRI data.

3.2. State space model(SSM)

Specifically, the Mamba series of algorithms primarily leverages the Selective Scan Space State Sequential Model (S6, Selective Scan Model) [10] to harness the sequential modeling capabilities of the State Space Model (SSM). Within the S6 block, a discretized SSM is utilized, which updates the hidden layer parameters through the framework of a Recurrent Neural Network (RNN). The state update computation process is as follows:

$$
\begin{aligned}
h'(t) &= A \cdot h(t) + B \cdot x(t) \\
y(t) &= C \cdot h(t) + D \cdot x(t)
\end{aligned}
\tag{1}
$$

where A is the state transition matrix, representing the evolution of the hidden state over time. B is the input influence matrix, projecting the input $x(t)$ into the hidden state space. C is the output matrix, transforming the hidden state $h(t)$ into the output. D is the direct influence matrix, directly affecting the output $y(t)$ through the input $x(t)$.

As shown in Fig. 2, the state update process can be intuitively represented: the hidden state is transformed by the state transition ($\times A$)

and input influence ($\times B$), then combined and passed through the output transformation ($\times C$), while the direct path ($\times D$) allows the input to directly contribute to the output.

These matrix parameters A, B, C, and D are learned during the training process. To incorporate the continuous-time SSM into a deep model, it must first be discretized. The basic idea is to solve the above ordinary differential equations, obtain the analytical solution, and then perform discretization. The specific formula is as follows:

$$h(t_b) = e^{A(t_b - t_a)} h(t_a) + e^{A(t_b - t_a)} \int_{t_a}^{t_b} B(\tau) u(\tau) e^{-A(\tau - t_a)} d\tau \tag{2}$$

After further discretization, the following equation is obtained:

$$h_b = e^{A(\Delta_a + \cdots + \Delta_{b-1})} \left(h_a + \sum_{i=a}^{b-1} B_i u_i e^{-A(\Delta_a + \cdots + \Delta_i)} \Delta_i \right) \tag{3}$$

To improve computational efficiency, the S6 block employs a Selective Scan mechanism, which effectively reduces computational complexity, enabling linear complexity calculations without losing the global receptive field.

Clearly, the majority of studies predominantly obtain feature sequences by segmenting individual images, which, although it allows for more complex temporal analysis after capturing spatial information, fails to assist the model in understanding and utilizing the dynamic changes between slices while recovering spatial features. Consequently, it overlooks the sequential associations within brain tumor MRI slices. To date, only a limited number of studies have focused on the image sequence features in semantic segmentation tasks [31–33].

4. Method

Given the aforementioned context, designing a sequential segmentation model tailored for MRI brain tumor data is a valuable endeavor. To address this, we have proposed an innovative semantic segmentation algorithm called SF-SSM UNet, which takes brain tumor MRI slices as image sequence inputs, and we have also formulated a training strategy suitable for models of this category.

4.1. Theoretical analysis

From a theoretical perspective, brain tumor MRI data exhibit not only high-dimensional spatial complexity but also sequential dependencies across slices. Let the input sequence be $X = \{x_1, x_2, \dots, x_T\}$, where each slice x_t represents one frame of an MRI volume. Conventional 2D segmentation models usually treat slices independently, assuming mutual independence among $P(x_t)$, which makes it difficult to capture inter-slice contextual dependencies. 3D convolutional models approximate the conditional distribution $P(x_t \mid x_{t-k}, \dots, x_{t+k})$ through local voxel operations; however, their receptive fields remain limited, and the computational cost is high, making it challenging to extend to long-range dependencies across the entire sequence.

The State Space Model (SSM) provides a more theoretically complete solution to this problem. SSM formalizes the input sequence using the state transition equations:

$$h_{t+1} = A h_t + B x_t, \quad y_t = C h_t + D x_t, \tag{4}$$

where h_t is the hidden state, A is the state transition matrix, B is the input projection matrix, C is the output mapping matrix, and D represents the direct input contribution. This formulation shows that the hidden state update depends on both the previous state and the current input, thereby establishing long-range dependencies in a linear recursive manner. Compared with recurrent neural networks (RNNs), SSM retains the recursive structure while alleviating the gradient vanishing problem; compared with the global attention mechanism in Transformers, SSM maintains a global receptive field while reducing the computational complexity to $O(T)$, making it more suitable for handling long MRI slice sequences.

In the proposed SF-SSM UNet, the sequential modeling module is embedded into both the encoder and decoder feature flows to capture global inter-slice dependencies. Specifically, the feature representation depends not only on the spatial information of a single slice $f(x_t)$ but also implicitly models the conditional distribution $f(x_t \mid x_{1:t-1})$ through state recursion, thereby unifying spatial features and temporal dependencies within the same representation space. In other words, the theoretical advantage of SF-SSM UNet lies in its ability to approximate the joint distribution $P(x_{1:T}, y)$ rather than the locally independent distribution $P(x_t, y)$. Therefore, the model can theoretically provide stronger representational power and better generalization, facilitating the capture of dynamic lesion evolution patterns across MRI sequences and improving the reliability of segmentation results.

4.2. SF-SSM UNet

The proposed SF-SSM UNet model primarily consists of the Sequential Flow State Space Model Group (SF-SSM Group) and the UNet backbone, which uses SS2D from V-Mamba as a traditional visual module, as illustrated in Fig. 3. The architecture of SF-SSM UNet follows the classic encoder–decoder structure and incorporates Skip-Connections. Both the encoder and decoder consist of three layers, each containing Vision Mamba's visual Selective Scan Module (SS2D, as shown in Fig. 3) and Patch up/down-sampling modules (bottom right of Fig. 3). An innovative SF-SSM Group (bottom left of Fig. 3) is integrated into each encoder layer. The SF-SSM Group serves as the core module of SF-SSM UNet, designed to capture sequential and channel associations within the image flow. Its key innovation lies in the cross-flowing mechanism, which alternates attention between the channel and sequential dimensions of the input image flow, enabling a cross-dimensional Selective Scan.

This model is capable of receiving and processing multi-channel images with sequential characteristics. Its input is defined as a set of multi-modal MRI image flows:

$$X = x_1, x_2, \dots, x_T \tag{5}$$

Where $x_t \in \mathbb{R}^{H \times W \times C}$ represents a single multi-modal MRI slice. $t \in [1, T]$ represents the number of frames in the image flow, i.e., the sequence length, where H, W, and C denote the height, width, and number of channels of the slice images, respectively. For the input image flow, SF-SSM UNet first performs image-level feature extraction through a Vision Mamba module (SS2D), followed by the proposed SF-SSM Group for attention perception at the temporal and channel levels.

4.3. Sequential flow selective scanning module group (SF-SSM group)

The proposed SF-SSM Group is the core module of SF-SSM UNet, as shown in Fig. 4. It is designed to enhance feature extraction along the sequence and channel dimensions. The group comprises four SF-SSM modules, each utilizing the Selective Scan State Space Sequential Model (S6) [10] as the core operator for processing image flows, and they are integrated through a Layer-Attention Module.

4.3.1. SF-SSM module

The SF-SSM applies the S6 module to the sequence and channel dimensions of the input image flow through a cross-flowing mechanism. When processing multi-channel MRI image flows, the SF-SSM selectively perceives variations along the sequence and channel dimensions, focusing the model's attention on the sequential flow information of lesions, as illustrated on the left side of Fig. 4. For each SF-SSM module, the frame count T of the input image sequential flow is first taken as the sequence length, and the number of channels C is taken as the model dimension, which is then input into the S6 Model for attention-aware

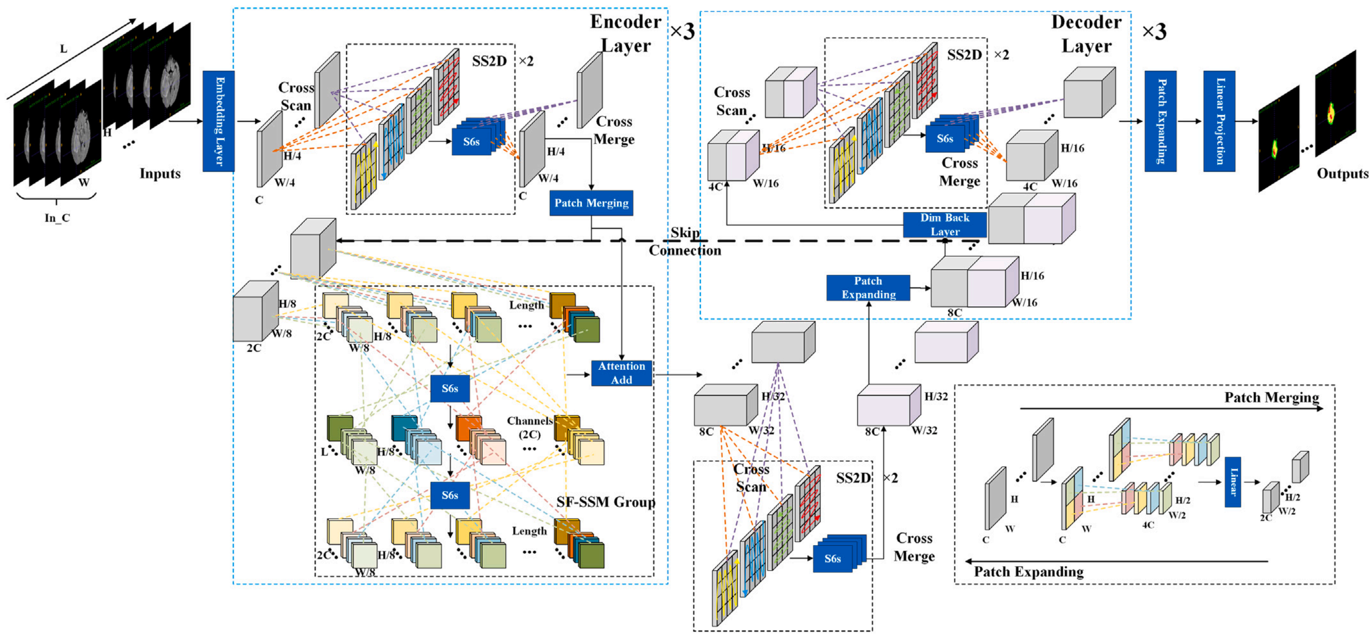

Fig. 3. Overall Architecture of SF-SSM UNet. It consists of three layers of a classic encoder-decoder framework, each containing a vision encoder based on Vmamba. The encoder also includes the core module SF-SSM Group for obtaining sequence associations.

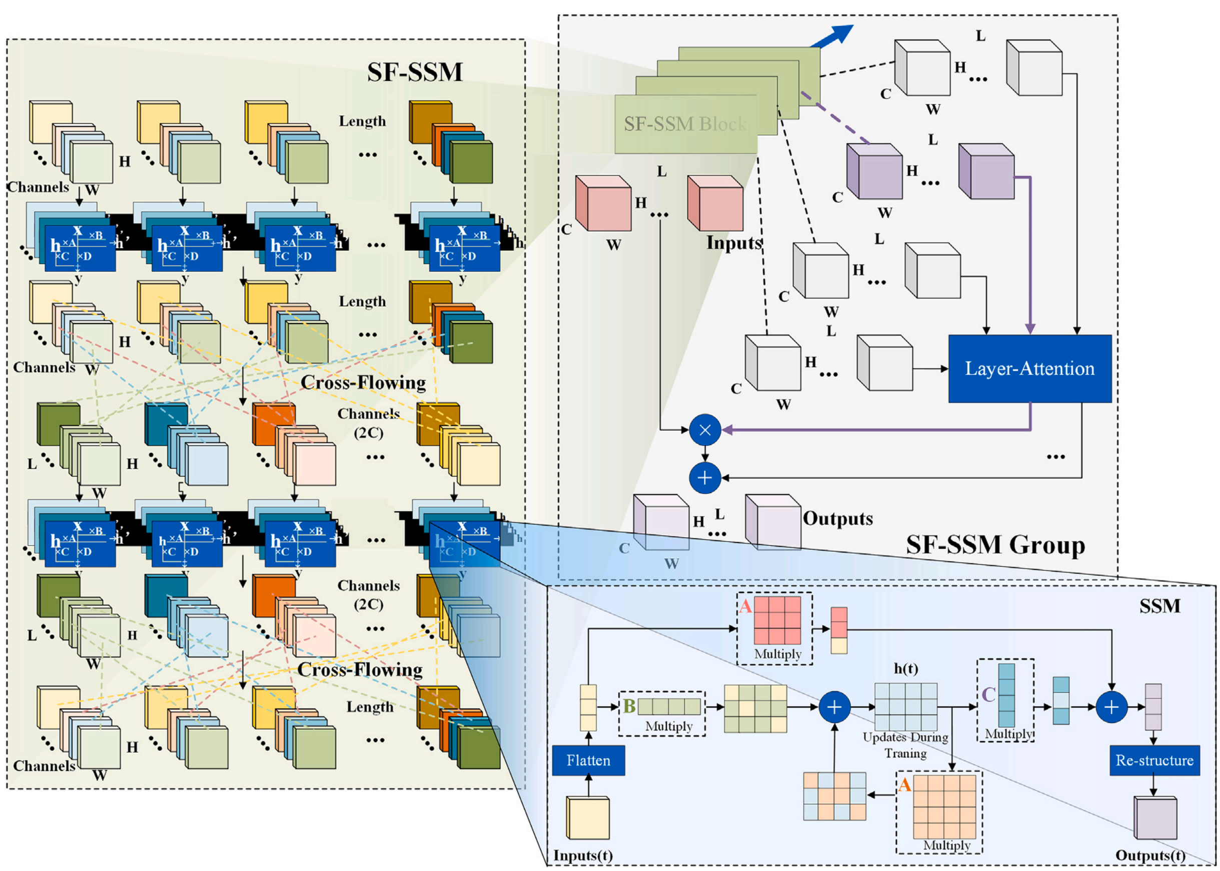

Fig. 4. Structural diagram of the core module SF-SSM Group. It consists of four SF-SSM modules based on S6 [10], which are used to extract sequence and channel correlations from image flows and are connected through layer attention.

processing. Here, the input sequential flow can be represented as:

$$X = \{x_1, x_2, \dots, x_T\}, \quad x_t \in \mathbb{R}^{H\times W\times C} \tag{6}$$

Let T represent the number of frames, i.e., the sequence length, and let H and W denote the height and width of each frame, respectively. C represents the number of channels (such as different signal channels in MRI images). The SF-SSM first feeds the input flow X into the S6 module. The S6 module is a state-space model similar to an RNN (as detailed in Section 2), designed for selective perception of temporal correlations along the sequence dimension. Since the S6 module can only accept one-dimensional sequential data as input, i.e., features with dimensions $T\times D$, the input $X_{in} \in \mathbb{R}^{T\times H\times W\times C}$ undergoes a certain dimensional reorganization to match the input requirements of the S6 while minimizing information loss, resulting in $X \in \mathbb{R}^{C\times T\times D}$ being propagated in the SF-SSM:

$$X = \{x_1, x_2, \dots, x_T\}, \quad x_t \in \mathbb{R}^{C\times D} \tag{7}$$

Where T is the sequence length from X_{in}, and D is the feature dimension obtained by flattening the slice, with $D = H \times W$. In the first layer of the SF-SSM process, the focus is more on sequence correlation, so the channel dimension C is temporarily set to the input BatchSize and does not participate in the computation.

In the S6 module, sequence feature correlations are selectively perceived through the state-space model. Each element interacts with previously scanned samples through compressed hidden states, enabling the model to understand frame-to-frame common correlations within the sequence, as illustrated at the bottom right of Fig. 4. For each frame x_t, the S6 module performs the following operations based on the state-space model:

State update equation:

$$h'(t) = A \cdot h(t) + B \cdot x(t) \tag{8}$$

Where $h(t)$ represents the hidden state at time t, denoting the model's memory. A and B are the state transition matrix and input influence matrix, respectively, which project the current frame $x(t)$ into the hidden state space.

Output equation:

$$y(t) = C \cdot h(t) + D \cdot x(t) \tag{9}$$

Where C and D are the hidden state output matrix and input projection matrix, respectively (distinct from the channel C and feature dimension D mentioned later), combining the hidden state $h(t)$ and input $x(t)$ to compute the output.

After each update, $h(t+1)$ is passed to the next time step to process the next input frame $x(t+1)$, continuing until all T frames have been processed. This ultimately produces an output sequence with the same dimensions as the input:

$$\hat{X} = S6(X), \quad \hat{X} \in \mathbb{R}^{C\times T\times D} \tag{10}$$

Subsequently, the input channel count C is treated as the sequence length, and the frame count T is treated as the model dimension for reorganization:

$$X' = \text{Transpose}(\hat{X}, (1, 0, 2)) \tag{11}$$

After reorganization, the channel count C is treated as the sequence length, and T is temporarily ignored as the BatchSize. The reorganized sequence is represented as:

$$X' = \{x'_1, x'_2, \dots, x'_C\}, \quad x'_c \in \mathbb{R}^{T\times D} \tag{12}$$

Next, the reorganized sequence is again fed into the S6 block for processing, yielding the output:

$$\hat{X}' = S6(X'), \quad \hat{X}' \in \mathbb{R}^{T\times C\times D} \tag{13}$$

Finally, the output is reorganized back to the initial dimensions to obtain the selectively scanned perceptual output along both the sequence and channel dimensions, which is then input into the next SF-SSM. Here, Re_View refers to restoring the image flow to its original input dimensions.

$$X_{out} = \text{Re\_View}(\text{Transpose}(\hat{X}', (1, 0, 2))), \quad X_{out} \in \mathbb{R}^{T\times H\times W\times C} \tag{14}$$

Overall, the algorithmic process of SF-SSM can be represented by Algorithm 1. By continuously inputting the dimension-restructured feature flows into the SSM and performing cross-flowing between modules, the spatiotemporal associations are effectively captured.

4.3.2. Layer-attention module

The outputs of the four internal modules within the SF-SSM Group are fused through concatenation, extracting generalized information about the lesion from the sequence and channel dimensions in the input stream. This information is progressively enhanced as the number of layers increases, resembling the layer-stacking process in RNNs. To maintain a balance among sequence, channel, and 2D semantic features, the model employs a Layer Attention mechanism. This mechanism directly multiplies the most balanced output feature with the input information, while adding the features from other layers as auxiliary information, as shown in the upper right of Fig. 4. The specific computation process is as follows:

Let the input to the i-th SF-SSM block be X_i, and its output be Y_i. The concatenation relationship can be expressed as:

$$Y_i = \text{SF-SSM}_i(X_i), \quad X_{i+1} = Y_i \tag{15}$$

For the first block, $X_1 = X_{\text{input}}$, which is the initial input data. The output Y_4 from the final block will contain the enhanced feature information after being processed through four layers of SF-SSM. As the number

Algorithm 1: SF-SSM Algorithm.

Input: $X_{in} \in \mathbb{R}^{T\times H\times W\times C}$: Input sequence,
  $S6$: State Space Model.
Output: $X_{out} \in \mathbb{R}^{T\times H\times W\times C}$: Output sequence after selective scanning.
$D \leftarrow H \times W$
$X \leftarrow Transpose(X_{in}, (3, 0, 1, 2))$ ▷ Reorganize dimensions to $C \times T \times H \times W$
$X \leftarrow View(X, (2, 3))$ ▷ Reorganize dimensions to $C \times T \times D$
$\hat{X} \leftarrow S6(X)$ ▷ Process sequence via the S6 module
$X' \leftarrow Transpose(\hat{X}, (1, 0, 2))$ ▷ Reorganize dimensions to $T \times C \times D$
$\hat{X}' \leftarrow S6(X')$ ▷ Process channel as sequence
$X_{out} \leftarrow Transpose(\hat{X}', (1, 0, 2))$ ▷ Reorganize dimensions to $C \times T \times D$
$X_{out} \leftarrow Re\_View(\hat{X}')$ ▷ Final output with original dimensions($T \times H \times W \times C$)
return X_{out}

of layers increases, the structure of the SF-SSM Group allows the model to progressively enhance its ability to capture lesion features in the input data.

For each layer's output feature Y_i ($i = 1, 2, 3, 4$), the importance of each layer's features is assessed by calculating the weights α_i through a hierarchical attention mechanism based on the SE (Squeeze-and-Excitation) Block:

$$\alpha_i = \mathrm{SE}(Y_i) \tag{16}$$

Where $\mathrm{SE}(\cdot)$ represents the Squeeze-and-Excitation operation. Subsequently, each layer's features are weighted according to α_i, with the most balanced feature Y_{balanced} being directly multiplied by the input information X_{input}, and the features from the other layers added as auxiliary information. Assuming Y_1 is the output of the layer with the highest weight:

$$Y_{\text{final}} = X_{\text{input}} \cdot Y_{\text{balanced}} + \sum_{i=2}^{4} \beta_i \cdot Y_i \tag{17}$$

Where β_i are the weighting coefficients for the auxiliary layer features, typically set to values less than 1 to avoid excessively amplifying non-primary features. The final output Y_{final} integrates features from different dimensions, ensuring a balance between sequence, channel, and 2D semantic features through the hierarchical attention mechanism.

4.4. Shuffled-ordered sequential flow training strategy(SOSF strategy)

The Frequency Principle [22], which posits that 'low frequencies first, followed by high frequencies, with high frequencies having a greater impact on details,' is widely recognized in deep learning. However, unlike one-dimensional data and two-dimensional images, it is challenging to define the frequency levels of sequential data. Here, we hypothesize that the frequency variation in sequence images is constituted by the inter-frame changes and the grayscale changes within the images themselves.

Due to the relatively fixed positions of organs and lesions in MRI, there is a certain correlation between non-continuous slices from the same case. It can be assumed that high-frequency sequence inputs (shuffled sequences) enable the model to learn more accurate sequence associations and lesion commonalities, while low-frequency sequence inputs (ordered sequences) allow the model's learning ability to focus on the high-frequency details of the images themselves. Experiments in Section 4.3.1 have confirmed the rationality of the aforementioned hypotheses and deductions. Based on this, we propose the Shuffled-Ordered Sequential Flow Training Strategy (SOSF Strategy) for the training of sequence models.

The SOSF Strategy is a two-phase training strategy aimed at enhancing the model's generalization ability, as illustrated in Fig. 5. This strategy introduces randomness and diversity, improving the model's adaptability to different sequential patterns while retaining its ability to understand and process the ordered sequence structure. This approach is particularly suitable for tasks requiring the simultaneous learning of 2D

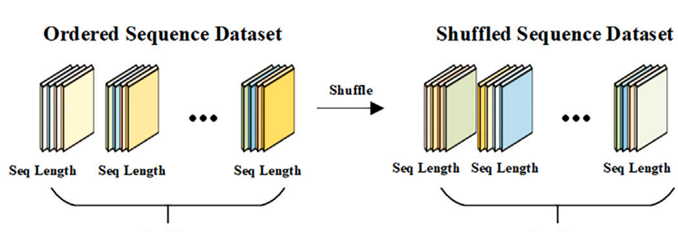

Fig. 5. Illustration of the SOSF Strategy. This approach enables the model to train separately on shuffled and Ordered sequence data, thereby learning both the common features of lesions and the unique characteristics of the sequence.

semantic and sequential features, such as the segmentation and analysis of brain tumor slices. The operational process is as follows:

Let the initial input X be a dataset consisting of n sequential image flows X_i, with each image flow X_i containing t elements. The initial input dataset $X = \{X_1, X_2, \ldots, X_n\}$ can be expressed as:

$$X_i = (x_{i1}, x_{i2}, \ldots, x_{it}) \tag{18}$$

In the Shuffled phase, each image flow element x_{ij} within the dataset is completely shuffled, generating a new Shuffled dataset X', where elements can originate from any image flow X_i and any position x_{ij}:

$$X' = \{X'_1, X'_2, \ldots, X'_n\} \quad \text{where} \quad X'_i = (x'_{i1}, x'_{i2}, \ldots, x'_{it}) \tag{19}$$

Here, x'_{ij} represents any shuffled image flow element from the ordered dataset. The elements of these sequences are globally shuffled, meaning that each new image flow may contain elements from any position within other sequences. By using the shuffled dataset X' as input for training, the model can better adapt to different input patterns and learn a broader range of features.

After training with the shuffled dataset, the model is then trained using the ordered sequence order. The goal of this phase is to fine-tune the model, ensuring it accurately understands and processes data arranged in the correct order.

The SOSF Strategy enables the model to sequentially capture both the common information of the lesion and the unique characteristics of specific image flow instances by leveraging the sequential flow summarization capabilities of the SF-SSM.

4.5. Loss function

The multi-class segmentation task for brain tumors can be converted into a multi-channel single-class segmentation task. During training, a combined loss function [23] using Binary Cross-Entropy Loss(BCE Loss) and Dice Loss is employed. BCE Loss is defined as follows:

$$L_{\text{bce}} = -\sum_{i=1}^{W}\sum_{j=1}^{H}\left[T_{ij}\log(P_{ij}) + (1 - T_{ij})\log(1 - P_{ij})\right] \tag{20}$$

where W and H represent the width and height of the predicted image $P(i, j)$ and the ground truth image $T(i, j)$.

The Dice Loss function [34] is specifically designed to handle class imbalance issues. In image segmentation tasks, the Dice Loss function can more accurately reflect the model's performance on small proportion lesion areas. Dice Loss can be expressed as:

$$L_{\text{dice}}(P, T) = 1 - \frac{2\sum_{i=1}^{N} p_i t_i + \tau}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} t_i + \tau} \tag{21}$$

where $p_i \in P$ represents the predicted image, $t_i \in T$ represents the ground truth image, τ is a small constant, and N is the number of pixels in the image.

Based on these two loss functions, a combined joint loss function is formulated as follows:

$$L_{\text{joint}} = \sum_{i=1}^{3}\left(\lambda L_{\text{dice}_i} + (1 - \lambda) L_{\text{bce}_i}\right) \tag{22}$$

where λ is the weight value for balancing different losses, with the range $0 < \lambda < 1$. To reduce the complexity of the task, we decompose the multi-lesion area segmentation task into a single-class segmentation task with multiple channels and calculate the losses separately. Here, L_{dice_i} and L_{bce_i} represent the Dice loss and cross-entropy loss for the i-th channel, respectively.

Table 2
Data number on brats-2019 dataset.

Method	Training sets (8 times)		Testing sets (2 times)
	Training	Valuation	Testing
Sequence Data	2483	275	702
2D Data	37,246	4139	10,540

Table 3
Data number on msd task01 dataset.

Method	Training sets (8 times)		Testing sets (2 times)
	Training	Valuation	Testing
Sequence Data	1295	143	359
2D Data	19,427	2158	15,396

5. Experiments

5.1. Dataset

BRATS-2019 [35–37] is a brain tumor dataset publicly released at MICCAI 2019. The training dataset includes 259 high-grade glioma (HGG) patients and 76 low-grade glioma (LGG) patients. Each MRI image has dimensions of 155 × 240 × 240. For training, we randomly selected 207 subjects from HGG and 60 subjects from LGG, and for testing, we randomly selected 52 subjects from HGG and 16 subjects from LGG, resulting in a training-to-testing ratio of 8:2. Each subject's data includes 4 MRI modalities (T1, T1c, T2, and FLAIR) and 4 labels (0 for non-tumor, 1 for necrosis and non-enhancing tumor, 2 for edema, 4 for enhancing tumor). The regions to be segmented include enhancing tumor (ET), tumor core (TC), and whole tumor (WT), with corresponding labels of 4, $1+4$, and $1+2+4$, respectively (Table 2).

MSD-Task01 (Brain Tumor) [53] is a brain tumor segmentation task publicly released as part of the Medical Segmentation Decathlon (MSD). This task contains multimodal MRI scans from 484 brain tumor patients, with each case providing four MRI modalities (T1, T1Gd, T2, and FLAIR) as well as corresponding voxel-level annotations. Unlike the BraTS dataset, MSD-Task01 labels the enhancing tumor (ET) as 3 instead of 4 (Table 3).

Considering that the black background in MRI images does not contribute to segmentation, and to address the imbalance between target and background data, the black background is cropped. After processing, each image is resized to 155 × 160 × 160. Subsequently, 155 two-dimensional slices of 160 $times$ 160 pixels were obtained through sectioning. All model algorithms will be trained using the same training set and evaluated on the same test set. For sequence segmentation models, the input to the model will be the slice sequences concatenated according to the sequence length. The specific data distribution is shown in Table 1. Additionally, since different modalities of MRI images have varying contrasts, Z-score normalization is applied to the foreground region of each image [38].

5.2. Metrics and implementation details

We implemented our network using the PyTorch framework on Ubuntu 22.04, with versions similar to those required by Mamba SSM. All experiments were conducted on an NVIDIA RTX 2080 Ti.

All algorithms were evaluated using Dice Score, Hausdorff95 Distance, Sensitivity, Positive Predictive Value(PPV) and Intersection over Union (IoU). Dice Score is a metric used to measure the similarity between two samples, especially in medical image segmentation to assess the consistency between predicted and ground truth segmentations. It is calculated as follows:

$$\text{Dice} = \frac{2 \cdot TP}{FP + 2 \cdot TP + FN} \tag{23}$$

where TP (True Positives) represents the number of correctly predicted positive samples, FP (False Positives) represents the number of incorrectly predicted positive samples, and FN (False Negatives) represents the number of positive samples that were not correctly predicted.

Hausdorff95 Distance is a metric for measuring the distance between two sets. It calculates the maximum of the minimum distances from each point in one set to the closest point in the other set. In the context of image segmentation, this metric measures the spatial discrepancy between the segmentation results and the ground truth annotations. It is computed as follows:

$$\text{Haus}(A, B) = \max\left(\max_{S_A \in S(A)} \left(d(S_A, S(B))\right), \max_{S_B \in S(B)} \left(d(S_B, S(A))\right)\right) \tag{24}$$

where A and B represent the sets of two segmentation results, d denotes the distance from an element in one set to the nearest point in the other set, and $S(A)$ and $S(B)$ represent the element sets of A and B, respectively.

In addition to Dice and Hausdorff95, we also report Sensitivity, Positive Predictive Value (PPV), and Intersection over Union (IoU) to provide a more comprehensive evaluation of segmentation performance. Sensitivity (also known as recall) measures the proportion of true tumor voxels correctly predicted, while PPV (precision) measures the proportion of predicted tumor voxels that are indeed correct. IoU evaluates the overlap between prediction and ground truth. Their definitions are as follows:

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \tag{25}$$

$$\text{PPV} = \frac{TP}{TP + FP}, \tag{26}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN}, \tag{27}$$

where TP, FP, and FN denote the number of true positive, false positive, and false negative voxels, respectively. High sensitivity indicates fewer missed tumor regions, high PPV indicates fewer false detections, and high IoU reflects better overall overlap between the predicted and reference regions.

5.3. Ablation study

5.3.1. Research on training strategy

Qualitative analysis. Typically, the input for sequence models consists of fixed-length sequential flow, which, compared to the shuffled regular 2D model inputs, lack randomness. According to the aforementioned hypothesis, using shuffled image flow inputs can be considered as sequential flow inputs with more high-frequency components, which would have a positive impact on the model's learning of sequence associations and commonalities in individual images. On the other hand, using shuffled image flow as inputs would prevent the model from learning the true characteristics of the lesion sequence, whereas ordered sequences, i.e., sequence inputs with more low-frequency components, would allow the model to learn more details of the images themselves.

We first verified that low-frequency image flow inputs (Ordered sequences) facilitate the model's learning of individual image details. We trained the SF-SSM UNet under different training strategies. In the Ordered sequences, which represent actual slice flows, the real sequential associations may positively influence the model's ability to learn detailed semantics in individual images. Fig. 6 provides examples of qualitative analysis, each example includes the Fourier transform spectrum of the ground truth image(Mask_fft), the spectrum of the predicted image(Pre_fft), their difference(Mask-Pre_fft), and the spatial domain difference obtained after applying the inverse Fourier transform to the difference(ifft). In the spectral difference maps, the model trained with Ordered sequences exhibited fewer errors in high-frequency regions. This is further supported by the edge information in the spatial domain images obtained after applying the inverse Fourier transform to the difference.

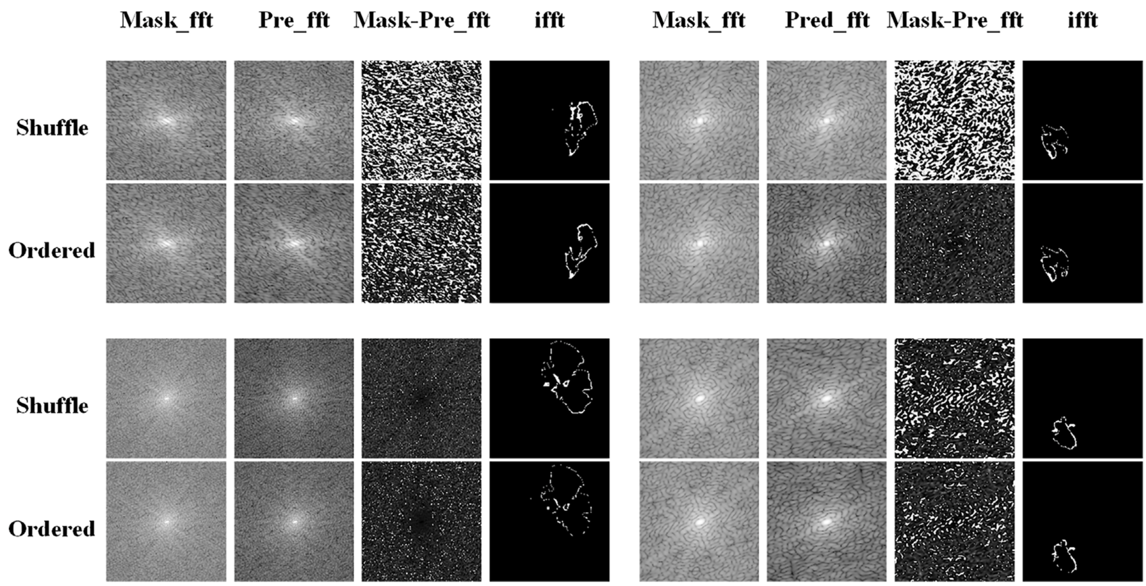

Fig. 6. Four examples of the visual comparison of high-frequency region errors in the test set for the SF-SSM UNet model trained with different strategies (Shuffled vs. Ordered).

Quantitative analysis. We observed that the frequency learning distribution of individual images differs when sequence models are trained with ordered sequences compared to unordered sequences. To quantify this difference and further explore the impact of these two strategies on sequential segmentation models, we define two metrics: the Frequency-domain Mean Error Ratio (FMER) and the Frequency-domain Mean Error Number (FMEN). Within a given frequency range, both metrics are used to quantify the spectral difference between the segmentation results and the ground-truth labels. FMER measures the average proportion of this error, while FMEN measures the average number of pixels with errors. Together, these two metrics enable the calculation and quantification of the error distribution within a specified frequency range.

Frequency-domain Mean Error Ratio (FMER) refers to the ratio of the average error value of each pixel within a certain frequency domain limit in the Fourier transform spectrum of all predicted results and true labels to the maximum error value. A lower FMER value indicates that the model's predictions in high-frequency detail regions are closer to the ground truth, resulting in smaller errors. Its calculation is as follows:

$$\mathrm{FMER} = \frac{1}{M}\sum_{j=1}^{M}\left(\frac{1}{N}\sum_{i=1}^{N}\left|F_{\mathrm{pred},ji}^{\mathrm{high}} - F_{\mathrm{label},ji}^{\mathrm{high}}\right|\right)\Big/ E_{\max} \tag{28}$$

where M represents the total number of images in the dataset, N represents the number of pixels in the high-frequency region, $F_{\mathrm{pred},ji}^{\mathrm{high}}$ denotes the value of the i-th pixel in the high-frequency region of the j-th predicted image's Fourier transform, $F_{\mathrm{label},ji}^{\mathrm{high}}$ denotes the value of the i-th pixel in the high-frequency region of the j-th true label image's Fourier transform, and $E_{\max}$ represents the maximum error value between the predicted result and the label image in the entire frequency spectrum, i.e., $\max\left(\left|F_{\mathrm{pred}} - F_{\mathrm{label}}\right|\right)$.

Frequency-domain Mean Error Number (FMEN) refers to the average number of error pixels within a certain frequency domain limit in the Fourier transform spectrum of all predicted results and true labels. A lower FMEN value suggests that the majority of pixels in high-frequency regions are predicted accurately, demonstrating better model performance in detailed areas. Its calculation is as follows:

$$\mathrm{FMEN} = \frac{1}{M}\sum_{j=1}^{M} P_{\mathrm{nonzero},j} \tag{29}$$

Table 4
Comparison of errors in the high-frequency region across different strategies. Red: best; bold black: second best.

High-Frq. Radius Ratio (r) (%)	Strategy	FMER(%)↓	FMEN↓
25	Shuffled	21.91	172.36
	Ordered	**21.75**	**171.48**
50	Shuffled	21.89	173.66
	Ordered	**21.73**	**170.76**
62.5	Shuffled	21.87	172.07
	Ordered	**21.71**	**170.64**
75	Shuffled	21.86	171.94
	Ordered	**21.69**	**170.51**

where M represents the total number of images in the dataset, and $P_{\mathrm{nonzero},j}$ denotes the number of pixels in the high-frequency region of the j-th image where $\left|F_{\mathrm{pred}}^{\mathrm{high}} - F_{\mathrm{label}}^{\mathrm{high}}\right| > 0$.

Together, FMER and FMEN provide a comprehensive reflection of the model's performance in high-frequency regions, specifically in the details and textures of the image. FMER reflects the intensity of the error, indicating the magnitude of the error in detailed regions, while FMEN reflects the extent of the error, indicating how many pixels in the detailed regions contain errors.

Table 4 shows the FMER and FMEN values of SF-SSM UNet trained on Ordered sequences and Shuffled sequences with different proportions of high-frequency regions in the WT label, where r represents the ratio of the low-frequency radius to the full image radius. A larger r value indicates that higher frequencies are considered within the computed region. The final results indicate that, both for FMER and FMEN, models trained on Ordered sequences achieve smaller errors in high-frequency regions. This suggests that the real sequential associations in Ordered sequences positively influence the model's learning of detailed semantics in individual images.

5.3.2. *Exploration of model architecture*

To analyze the role of the sequence and channel components in the proposed SF-SSM Group and the impact of different training strategies on performance, we designed ablation experiments using the U-shaped network (Mamba UNet) with V-Mamba as the base encoding module as

Table 5
Ablation study on brats-2019 dataset. Red: best; bold black: second best.

Model and Method	Dice_score(%)			Haussdorff95		
	WT↑	TC↑	ET↑	WT↓	TC↓	ET↓
BaseLine (Ordered)	87.17	89.29	90.41	1.3710	0.8875	0.7093
SF-SSM UNet (S, Ordered)	87.86	89.28	90.93	1.3424	0.8736	0.6895
SF-SMM UNet (S+C, Ordered)	88.05	90.21	90.65	1.3332	0.8465	0.7064
BaseLine (Shuffled)	**88.21**	90.11	90.86	**1.3061**	0.8235	0.6750
SF-SSM UNet (S+C, Shuffled)	88.07	**90.32**	**91.05**	1.3114	**0.8169**	**0.6719**
SF-SSM UNet (S+C, SOSF Strategy)	**88.45**	**90.55**	**91.44**	**1.2835**	**0.8152**	**0.6569**

the BaseLine, as shown in Table 5. Among them, red indicates the best result, and bold black indicates the second best. Here, S and C represent the sequence and channel associations focused on in SF-SSM, respectively.

When using Ordered image flow inputs, the SF-SSM Group, which includes both sequence and channel information, achieved the best results compared to the BaseLine and the sequence SSM alone. The experimental results using Shuffled image flow inputs also indicate that using the SF-SSM Group in the encoder yields better performance than not using it. This suggests that employing the SF-SSM Group with sequence and channel dimension information in the encoder part is optimal, which will also serve as the final configuration for SF-SSM UNet.

On the other hand, comparing the same configuration of SF-SSM UNet, it can be observed that the performance on Shuffled sequences is somewhat superior to that on Ordered sequences. Combining the positive impact of Ordered sequences on image detail areas, as discussed earlier, it can be concluded that integrating the advantages of both is an effective training strategy. The experimental results comparison in the same configuration of SF-SSM UNet also indicate that the model trained with the SOSF Strategy achieved the best performance.

5.4. Comparison with the state-of-the-art methods

Tables 6 and 7 presents the comparison results of the proposed SF-SSM UNet with other medical image segmentation methods on the BRATS-2019 dataset. The compared methods include U-Net [4], SegResNet [39], Cascaded UNet [46], TransUNet [24], UNETR [26], SwinUNETR [27], MedNeXt [28], DconnNet [40], MedSAM [20], Mamba-UNet [30], U-KAN [41], VM-UNet [42], MambaBTS [47], Mamba Sea [43], UD-Mamba [44], and H-VMUNet [45].

To ensure a fair comparison, all algorithms were evaluated on the same dataset using the same loss function, with training conducted for up to 300 epochs and early stopping applied after 30 epochs. The best results are highlighted in red bold, while the second-best results are shown in black bold. For some recently proposed methods without publicly available code, we were unable to retrain them under the same settings to obtain test results and runtime. Therefore, their official results reported on the BraTS-2019 are included in the comparison and marked with an asterisk (*).

The metrics in Tables 6 and 7 indicate that the proposed SF-SSM UNet outperforms all other algorithms across all metrics for each region, demonstrating a significant performance lead. Specifically, the SF-SSM UNet achieves Dice scores of 88.45, 90.55, and 91.44, and Haussdorff95 scores of 1.2835, 0.8152, and 0.6569 for the WT, TC, and ET regions, respectively. The proposed method also achieves the best performance in terms of sensitivity, PPV, and IoU, representing the current state-of-the-art (SOTA).

In addition, Fig. 7 presents qualitative visualization results comparing the proposed SF-SSM UNet with other representative methods. The green boxes indicate false detections, where non-tumor regions are incorrectly identified as tumor areas. It is evident that the proposed method provides more accurate delineation of tumor boundaries and effectively reduces false-positive regions. Meanwhile, SF-SSM UNet better preserves fine-grained structures within tumor areas, capturing subtle lesion characteristics and thereby improving the overall reliability of the segmentation results. This advantage is consistent with the quantitative improvements reported in the tables, further validating the superiority of the proposed method in terms of accuracy and detail fidelity.

Tables 8 and 9 present the comparative experiments of all the above algorithms on the MSD Task01 dataset. Similar to the results on BraTS-2019, the proposed SF-SSM UNet achieves a significant lead in the primary metrics of Dice and Hausdorff95 distance. The Dice scores for the WT, TC, and ET regions are 89.26, 88.74, and 90.35, respectively, while the corresponding Hausdorff95 distances are 1.3291, 0.9256, and 0.7701. Most of the metrics achieve either the best or second-best rankings.

Table 6
Comparison with the sota methods on brats-2019 (dice & Haussdorff95). Red: best; bold black: second best.

Model	Year	Dice_score(%)			Haussdorff95		
		WT↑	TC↑	ET↑	WT↓	TC↓	ET↓
UNet	2015	87.36	88.59	90.69	1.3582	0.9076	0.6897
SegResNet	2019	87.89	89.58	91.14	**1.2977**	0.8403	0.6649
Cascaded UNet	2019	87.81	89.40	90.92	1.3349	0.9002	0.6719
TransUNet	2021	84.50	86.72	88.39	1.3911	0.9300	0.7396
UNETR	2022	85.29	87.16	89.54	1.3831	0.9504	0.7042
Swin UNETR	2022	88.16	88.85	90.86	1.3077	0.9119	0.6814
MedNeXt	2023	87.55	89.18	90.45	1.3330	0.8800	0.6958
DconnNet	2023	87.08	89.12	90.57	1.3093	0.8390	0.6758
MedSAM	2024	85.39	87.90	88.20	1.4409	0.9224	0.7667
Mamba-UNet	2024	**88.21**	90.11	90.86	1.3061	**0.8235**	0.6750
UKAN	2024	87.39	89.50	**91.20**	1.2989	0.8415	**0.6582**
VM-UNet	2024	87.74	**90.39**	91.06	1.3122	0.8258	0.6744
Mamba BTS*	2024	84.50	86.06	77.96	2.6511	1.6086	2.7813
UD-Mamba	2025	88.03	89.85	90.05	1.3053	0.8634	0.7021
H-VMUNet	2025	87.95	90.03	90.50	1.3104	0.8411	0.6821
Mamba Sea	2025	88.19	90.19	90.78	1.3081	0.8385	0.6818
SF-SSM UNet (ours)	–	**88.45**	**90.55**	**91.44**	**1.2835**	**0.8152**	**0.6569**

Table 7
Comparison with the sota methods on brats-2019 (Sensitivity, PPV & IoU). Red: best; bold black: second best.

Model	Year	Sensitivity(%)			PPV(%)			IoU(%)		
		WT↑	TC↑	ET↑	WT↑	TC↑	ET↑	WT↑	TC↑	ET↑
UNet	2015	90.01	93.35	93.39	92.68	91.73	92.58	77.56	79.52	82.97
SegResNet	2019	90.23	92.43	93.17	92.24	93.04	92.59	78.40	81.13	83.72
Cascaded Net	2019	90.45	93.94	93.41	93.53	92.02	92.81	78.27	80.83	83.35
TransUNet	2021	87.61	91.05	91.58	91.97	91.75	92.84	73.16	76.55	79.20
UNETR	2022	88.38	89.90	91.36	92.00	93.00	93.06	75.89	77.24	81.06
Swin UNETR	2022	89.09	91.81	93.23	93.05	**94.34**	93.84	78.83	79.94	83.25
MedNeXt	2023	90.63	92.69	93.28	92.57	93.09	93.75	77.86	80.47	82.57
DconnNet	2023	88.39	91.81	92.32	**94.00**	93.52	93.20	77.12	80.38	82.77
MedSAM	2024	86.84	90.29	90.47	93.49	93.62	93.10	74.50	78.41	78.89
Mamba-UNet	2024	90.65	93.28	93.92	93.22	93.79	93.87	**78.91**	82.00	83.25
UKAN	2024	88.67	91.24	92.11	93.55	93.68	**94.20**	77.60	81.00	**83.82**
VM-UNet	2024	**91.11**	**94.34**	**94.27**	93.58	**94.14**	92.72	77.16	80.53	82.88
Mamba BTS*	2024	87.16	90.62	83.05	85.97	89.20	78.94	–	–	–
UD-Mamba	2025	90.81	93.91	93.75	93.32	93.28	93.80	78.62	81.57	81.9
H-VMUNet	2025	90.52	92.23	94.22	93.71	93.98	93.28	78.49	81.87	82.65
Mamba Sea	2025	90.76	93.62	93.38	93.29	93.37	94.06	78.87	**82.13**	83.12
SF-SSM UNet(ours)	–	**91.21**	**94.92**	**94.46**	**94.07**	94.11	**94.07**	**79.32**	**82.72**	**84.22**

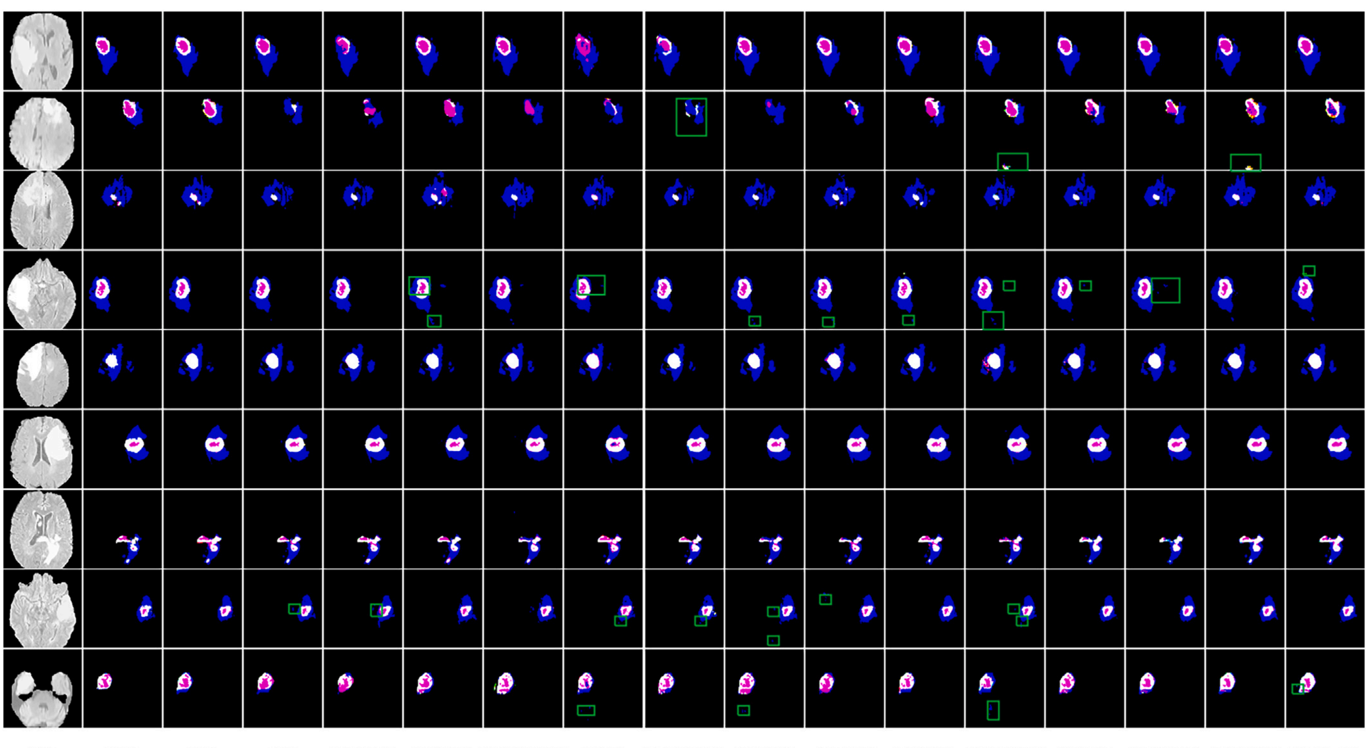

Fig. 7. Examples of Multi-method segmentation results. For the segmentation result image, where Blue: peritumoral edematous/invaded tissue, red: necrotic tumor core, White: GD-enhancing tumor.

Table 10 presents the computational efficiency analysis of SF-SSM UNet compared with current mainstream algorithms, where the input for all methods is standardized to a batch size of 15. Compared with other Mamba-based algorithms, SF-SSM UNet exhibits a similar number of parameters and FLOPs, with the additional overhead mainly attributed to the sequential modeling module designed for sequence inputs. Compared with other approaches, SF-SSM UNet achieves a lower computational burden and faster inference speed, demonstrating that incorporating sequence modeling does not introduce substantial computational cost while significantly improving model performance.

5.5. Statistical significance analysis

To further validate the superiority of the proposed method, we conducted statistical analyses based on the experimental results on the BraTS-2019 dataset. Specifically, six evaluation tasks (Dice and Hausdorff95 for WT, TC, and ET) were used for task-level statistical tests across different models. First, for each task we compared the performance of SF-SSM UNet against baseline methods and recorded the win–loss counts. The results showed that SF-SSM UNet consistently achieved advantages of either 6 wins and 0 losses or 5 wins and 1 loss when compared with strong baselines such as Mamba-UNet, UD-Mamba, H-VMUNet, and Mamba

Table 8
Comparison with the sota methods on msd task01 (Dice & Haussdorff95). Red: best; bold black: second best.

Model	Year	Dice_score(%)			Haussdorff95		
		WT↑	TC↑	ET↑	WT↓	TC↓	ET↓
UNet	2015	88.15	88.56	90.33	1.3800	0.9356	0.7861
SegResNet	2019	89.08	88.52	90.07	1.3562	0.9440	0.7824
Cascaded UNet	2019	89.06	88.29	90.24	1.3510	**0.9268**	0.7648
TransUNet	2021	87.26	88.58	89.52	1.3936	0.9349	0.8073
UNETR	2022	85.92	85.82	88.52	1.4480	1.0447	0.8310
Swin UNETR	2022	89.20	88.51	89.97	1.3571	0.9372	0.7989
MedNeXt	2023	88.41	87.80	89.56	1.3913	0.9525	0.8139
DconnNet	2023	88.46	88.46	90.12	1.3787	0.9374	0.7952
MedSAM	2024	84.20	86.11	86.72	1.5697	1.0025	0.9206
Mamba-UNet	2024	88.75	88.38	89.90	1.3459	0.9280	0.7879
UKAN	2024	87.81	88.10	**90.35**	1.3410	0.9355	0.7880
VM-UNet	2024	89.05	88.43	89.76	1.3393	0.9250	0.7912
UD-Mamba	2025	88.81	88.24	89.83	1.3408	0.9320	0.7895
H-VMUNet	2025	**89.12**	**88.59**	90.14	1.3341	**0.9244**	0.7789
Mamba Sea	2025	**89.15**	88.40	90.28	1.3398	0.9352	**0.7735**
SF-SSM UNet (ours)	–	**89.26**	**88.75**	**90.35**	**1.3291**	0.9256	**0.7701**

Table 9
Comparison with the sota methods on msd task01 (sensitivity, PPV & IoU). Red: best; bold black: second best.

Model	Year	Sensitivity(%)			PPV(%)			IoU(%)		
		WT↑	TC↑	ET↑	WT↑	TC↑	ET↑	WT↑	TC↑	ET↑
UNet	2015	90.71	91.46	92.97	93.11	93.11	93.60	78.81	79.47	82.37
SegResNet	2019	**92.66**	92.36	93.52	92.12	91.40	92.63	80.31	79.40	81.93
Cascaded Net	2019	91.25	91.93	93.09	93.50	**93.39**	93.84	80.18	79.35	81.88
TransUNet	2021	90.40	91.58	92.30	92.04	92.52	92.92	77.40	79.50	81.03
UNETR	2022	89.01	89.12	92.18	92.03	91.34	91.82	75.32	75.16	79.40
Swin UNETR	2022	**92.86**	**92.60**	**94.51**	91.83	91.54	91.43	**80.51**	79.39	81.77
MedNeXt	2023	89.39	91.22	93.78	92.16	92.25	91.69	79.23	78.25	81.09
DconnNet	2023	91.54	91.90	93.39	93.25	92.75	93.33	79.31	79.31	82.02
MedSAM	2024	88.07	88.46	91.36	90.75	**93.60**	90.32	72.71	75.61	76.55
Mamba-UNet	2024	90.41	91.70	93.59	**94.08**	91.96	91.91	79.78	79.18	81.65
UKAN	2024	89.75	92.43	**93.85**	93.66	93.10	92.55	78.27	78.73	**82.40**
VM-UNet	2024	90.68	92.40	93.33	93.51	91.37	91.35	**80.26**	79.26	81.42
UD-Mamba	2025	90.75	92.34	93.47	93.13	91.25	92.05	79.87	78.95	81.54
H-VMUNet	2025	90.63	**92.55**	93.62	93.20	91.47	92.32	80.38	**79.52**	82.05
Mamba Sea	2025	91.32	92.35	93.56	93.14	91.26	92.40	**80.42**	79.21	**82.28**
SF-SSM UNet(ours)	–	91.45	**92.60**	**94.03**	**94.17**	**93.11**	**93.99**	**80.60**	**79.78**	**82.40**

Table 10
Comparison of model parameters and computational complexity.

Model	Year	Params (M)	FLOPs (G)	Inf Time (min:second)
UNet	2015	39.40 M	321.19 G	12:32
SegResNet	2019	92.59 M	5.98 G	10:54
Cascaded UNet	2019	85.59 M	568.25 G	25:24
TransUNet	2021	105.21 M	237.83 G	11:02
UNETR	2022	149.13 M	150.71 G	18:31
Swin UNETR	2022	69.14 M	136.80 G	21:33
MedNeXt	2023	3.30 M	1.98 G	29:42
DconnNet	2023	212.39 M	128.55 G	20:34
MedSAM	2024	240.32 M	166.55 G	30:19
Mamba-UNet	2024	239.47 M	72.44 G	14:12
UKAN	2024	25.36 M	62.21 G	19:43
VM-UNet	2024	44.28 M	61.42 G	13:52
Mamba BTS*	2024	–	–	–
UD-Mamba	2025	39.49 M	63.29 G	14:35
H-VMUNet	2025	51.92 M	75.98 G	16:04
Mamba Sea	2025	27.43 M	66.91 G	16:49
SF-SSM UNet (ours)	–	81.59 M	85.51 G	14:47

Sea. The sign test indicated that these advantages were statistically significant (< 0.05, remaining significant after Holm–Bonferroni correction).

Furthermore, we converted the performance of all methods on the six tasks into ranks and applied the Friedman test for an overall comparison. The results demonstrated significant differences among the methods ($p < 0.01$). A subsequent Nemenyi post-hoc comparison revealed that the average rank of SF-SSM UNet was significantly better than that of most competing methods, particularly Mamba-UNet, Swin UNETR, UNETR, and MedSAM ($p < 0.05$).

In summary, the statistical analysis not only confirms the consistent advantages of the proposed method in individual tasks, but also demonstrates that SF-SSM UNet achieves statistically significant improvements over existing methods in the overall comparison.

6. Discussion and conclusion

Accurate segmentation of brain tumor magnetic resonance imaging(MRI) images is critical for disease diagnosis and treatment. However, existing algorithms often overlook the sequential characteristics of brain tumor MRI slices and fail to fully exploit their sequence associations, which limits their performance in precise tumor segmentation tasks. To address this limitation, this study moves beyond the constraints of traditional two-dimensional or three-dimensional models by treating brain tumor MRI slices as a serialized image flow. The proposed SF-SSM UNet algorithm, along with its core mechanism, the SF-SSM Group, applies the Mamba SSM to the sequence and channel dimensions of the input flow through cross-flowing operations, thereby enabling the model to effectively learn sequence-flow-related features from both the sequence and feature dimensions.

Additionally, based on the Frequency Principle, this study introduces the SOSF strategy as a training method for sequential models to enhance their ability to learn features from sequence flows. The experimental results obtained on the BRATS-2019 dataset demonstrate that the proposed method achieves significant improvements in segmentation accuracy and visual quality compared to existing algorithms, achieving state-of-the-art performance.

This validates the feasibility and effectiveness of sequence flow segmentation for semantic segmentation tasks in medical imaging. Future work will further investigate the sequential characteristics of medical images, such as MRI, and develop more precise and generalized algorithms.

Although the proposed SF-SSM UNet achieves excellent performance on datasets such as BraTS-2019 and MSD Task01, several limitations remain. The most critical one is that SF-SSM UNet is specifically designed for brain MRI data, where slice-to-slice correlations are relatively clear and brain size variations are limited. In contrast, for other lesion imaging modalities such as lung CT, where organs and lesions exhibit more dramatic size variations, capturing sequential correlations may be more challenging. Thus, the generalization ability of SF-SSM UNet to other types of lesions still needs to be further improved.

In the future, we plan to further investigate the sequential characteristics of various medical imaging modalities and extend this modeling strategy to segmentation tasks of more organs and lesions, and potentially beyond segmentation tasks alone.

CRediT authorship contribution statement

Jiacheng Lu: Writing – original draft, Visualization, Validation, Software, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Hui Ding:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Qirun Huo:** Writing – review & editing, Validation, Resources. **Kaiwen Wang:** Validation, Project administration, Investigation. **Xinyu Sun:** Validation, Methodology, Investigation. **Shiyu Zhang:** Validation, Investigation, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Hui Ding has patent pending to Capital Normal University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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